



Urban environmental inequalities in Latin America: A scoping review

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ABSTRACT

Latin America is among the most urbanized regions in the world. However, cities in this region usually present an uneven distribution of environmental amenities and hazards between socioeconomic sectors, translating into prevalent issues of environmental inequalities. Urban environmental inequalities have been widely covered by studies from developed regions, but less attention has been placed on highly urbanized regions from the developing world. In this work, we review the topic of urban environmental inequalities in Latin America, aiming to identify the temporal and geographical trends of published research, and revise the main themes framing the research on environmental inequalities in this region. We searched for literature in three databases (i.e. Scielo, Scopus, Web of Science) using three different languages (i.e. English, Spanish, Portuguese). We identified a total of 57 articles published on the topic. These articles show an increasing rate of publication and citations during the last decade, with increasing proportion of papers published in English in journals covered by international databases. There is a large geographical bias towards a few countries and cities, and more than half of Latin-American countries have no research on this topic yet. While in North America race and ethnicity have been often associated to urban environmental inequalities, in Latin America most of studies relate environmental inequalities to social and economic variables. These results highlight the relevance of carrying out this type of research in developing regions, as these may provide key novel insights for generating better solutions to address urban environmental inequalities.

1. Introduction

We are living in the urban era. Year 2007 marked a key point in human civilization, as for the first time in history, more than 50% of human population lived in urban areas. Currently, close to 57% of the global estimated population lives in urban areas, and it is expected that this number increases to 68% by the middle of the century [91]. The large proportion of this urbanization is happening in developing regions, with millions of people leaving their rural live to migrate to cities searching for opportunities to achieve a better quality of life [86]. Nevertheless, while urbanization has shown to promote social, technological and economic development, it has also shown to have produced a diversity of urban issues, such as environmental degradation, social segregation and increasing levels of economic inequalities [61,86].

Latin America is among the regions with the highest rates of urbanization worldwide, with over 81% of the population living in urban areas, and close to 50% of the population living in cities larger than 1 million inhabitants [91]. However, the rapid urbanization growth experienced by Latin-American countries during the last decades has not been linked to proper urbanization policies, which has been translated in cities with high levels of socioeconomic segregation and large inequalities

in the accessibility to urban amenities [92]. The frequent spread of informal settlements in urban peripheries has produced a direct loss and degradation of natural habitats but has also pushed Latin-American governments to allow the expansion of cities in areas prone to environmental risks, and in some cases to regularize settlements in areas lacking key urban infrastructures such as piped water and sewage collection systems [30]. Furthermore, the combination of economic inequalities and socioeconomic segregation has produced large disparities in the spatial distribution of environmental amenities and hazards, generating cities with high levels of urban environmental inequalities (e.g. [32,51,85,97]).

Environmental inequality refers to the “unequal social distribution of environmental risks and hazards and access to environmental goods and services” [81]. Environmental inequalities do not necessarily imply that the underlying process or the resulting distribution of the environmental factor is unfair, but rather refer to a distributional pattern that may be indicative of environmental justice issues [32]. Although closely related, environmental justice is a broader theme, that not only focuses on the unfair distributional patterns of environmental factors, but also on other related dimensions such as power relations, institutional drivers, social movements, and correcting practices [2]. Thus, while urban en-

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environmental justice studies imply the existence of unfair environmental inequalities, urban environmental inequality studies may not necessarily result in environmental justice issues. For example, in the city of Santo Domingo (Dominican Republic) there is important inequalities in the distribution of tree cover between socioeconomic groups. Yet, the lower-income sectors are the one showing the largest proportion of tree cover [55], which can be hardly classified as an environmental injustice.

Although the study of urban environmental inequalities has experienced a wide coverage during the last decades, the large number of studies on this topic comes from developed regions [82], leaving a gap of knowledge regarding the state of the art on this topic for highly urbanized developing regions, such as Latin-American. In addition, as urban form and urban development trajectories may greatly differ between developing and developed regions [45], the diversity and characteristics of environmental inequalities reported on the mainstream literature may be a biased representation of the environmental, social and economic factors associated to environmental inequalities in other parts of the world. This could be even more relevant for non-English speaking regions, as many studies could be published in local or regional journals using local languages, which may not be indexed in main scientific databases, and may often go unnoticed due to the use of non-English language [47,48,50]. In this regard, developing studies to synthesize the advances on urban environmental inequality studies in Latin America could be of great importance to understand how this topic has been developed in this region, but may also help to reduce the scientific bias regarding the accumulated knowledge in this area.

Currently, there are no reviews covering urban environmental inequalities in Latin America published in English or in journals covered by main scientific database (i.e. Web of Science [WoS], Scopus). However, a review on the related topic of urban environmental justice based on grey and local scientific databases can be found in a local journal published in Spanish [56]. Unfortunately, this review has received almost no scientific attention so far (Google Scholar shows only one citation), highlighting the relevance of developing efforts to publish this type of reviews in English language in journals covered by mainstream scientific databases. In addition, while including non-peer-reviewed documents and local databases may be useful for a deeper understanding of emerging themes [54,68], it usually faces the challenge of screening the quality of published documents, which may incorporate researcher's filtering bias, therefore reducing the comparability of results with other studies. Thus, carrying out a literature review based on peer-review journals and well-established scientific databases may help to further understand the trends of scientific publications on the topic of urban environmental inequality in Latin America, also providing a showcase platform for those relevant studies that may have gone less noticed due to language barriers. Of course, this will require to reduce language bias by also including papers published in the prevailing local languages [58].

In this work we performed a scoping review [68] to assess the current state of knowledge regarding urban environmental inequalities in Latin America and to identify the main temporal, geographical and thematic patterns of published research. We searched for peer-reviewed papers published between 1980-2020 in three different languages (i.e. English, Portuguese, Spanish) in three main scientific databases (i.e. Scielo, Scopus, WoS) aiming to: (1) identify the temporal trends of published research on the topic; (2) evaluate the geographical distribution of studied areas and countries developing the studies; and (3) revise the main social and environmental themes framing environmental inequalities research in Latin America.

2. Methodology

2.1. Literature search, screening and depuration

To reduce language, and language-indexing bias, we performed the publications search in three languages (i.e. English, Portuguese, Spanish) using two well-known scientific data bases (i.e. Scopus, WoS

Core Collection) and the largest Latin-American bibliographic database, named Scielo (Scientific Electronic Library Online). The search in all platforms were performed on August 10, 2021. In all databases we applied the same filters for searching; (1) timespan: 1980 – 2020; (2) search in title, abstract and keywords; (3) research and review articles only, and (4) English, Spanish and Portuguese language only. We used a long timespan aiming to detect the first papers published on the topic, assuming that due to the rise of the environmental justice studies in the global North by the beginning of the 80's decade [59], it would be difficult to have registries of papers with a focus on Latin America before 1980.

We applied equivalent search syntax in the three databases, using an independent search for each of the three languages. For example, the search syntax used for English language in WoS was as follow: ("Environment* Inequ*" OR "Environment* *justice") AND (*urban* OR city OR cities) AND (Argentina OR Bolivia OR Brazil OR Chile OR Colombia OR "Costa Rica" OR Cuba OR "Dominican Republic" OR Ecuador OR "El Salvador" OR Guatemala OR Honduras OR Mexico OR Nicaragua OR Panama OR Paraguay OR Peru OR Uruguay OR Venezuela). The specific syntaxes used in the other languages and databases are detailed on the supplementary material.

Using this search strategy, we obtained a total of 371 registries adding the results from the three databases and languages (Fig. 1). The search on the WoS database only produced results for the English query, with 82 registries. Scopus resulted in 98 documents for the English, 4 for Spanish and 1 for the Portuguese syntax search. Scielo produced the largest number of results, with 108, 48 and 30 registries for the English, Spanish and Portuguese query, respectively. A total of 142 registries were removed because were duplicated within or between databases, leaving a list of 229 individual articles for screening.

Because the search strategy could have included articles beyond the scope of our analysis, such as studies not focused on urban environmental inequalities or covering urban areas located outside the Latin-American Region, each of the three authors independently performed an in-depth screening (i.e. full text reading) of the 229 articles by evaluating if the articles followed these seven rules: (1) Is the article published between 1980 and 2020? (2) Does the article fulfill the search strategy within the title, abstract and keywords? (3) Is the main body of the article written in English, Spanish or Portuguese? (4) Is the article a review or research article (not an editorial, short communication or opinion piece)? (5) Is the article covering urban or periurban areas? (6) Are these urban areas located in a Latin-American country? (7) Is the article main objective covering the topic of environmental inequality or justice? To perform this task, each author had an individual spreadsheet to classify each of the 229 articles based on the above rules into one of three categories: include, exclude, undecided. After individual classification we integrated our spreadsheets and identified all the articles that we had not 100% of agreement. Then we set specific working sessions to further screening those articles until the three authors agreed on the final classification. After this procedure we excluded 172 articles, ending with a total of 57 articles for the analysis (Fig. 1).

2.2. Database completion and standardization

Because the 57 selected papers were collected from different databases, we proceeded to complete and standardize the data to have the exact format and type of information in our database (following the WoS database format), in particular for those fields that were key for our analysis, such as the article language, abstract and country corresponding to the institution of affiliation of the corresponding author.

2.3. Temporal and geographical patterns

We built a secondary database with the number of citations per year of each paper. We did this based on the information available on the articles' profile in Google Scholar. For those papers we could not find the

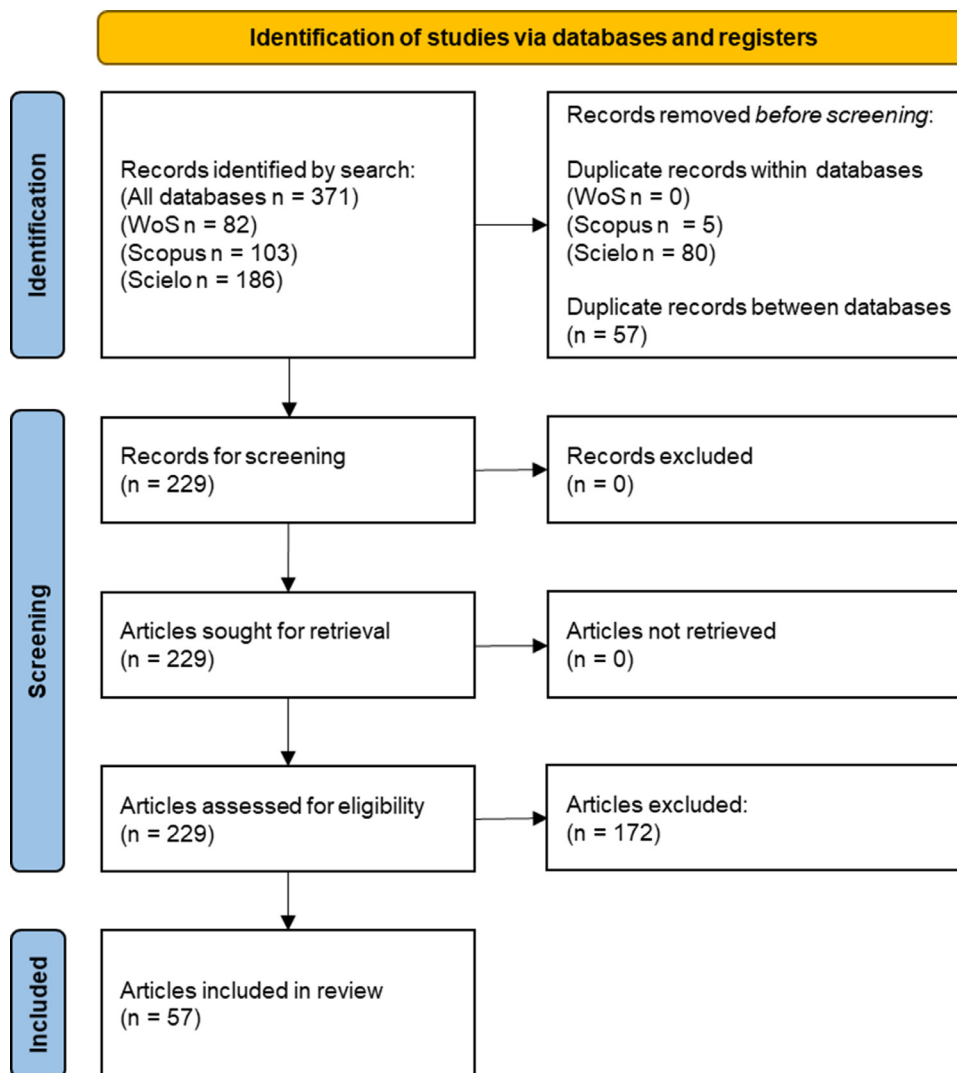


Fig. 1. Flowchart of the protocol used in this systematic review based on the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) [66].

article profile, we manually counted the number of citations per year using the “Cited by” option on Google Scholar, counting only the documents showing publication year. We built another database with the specific urban areas covered by the articles, their respective georeferenced coordinates and the number of articles covering those areas.

2.4. Thematic analysis

We used the wordcloud (v. 26) package in R software to generate a wordcloud depicting the 30 most frequent words contained in the abstracts of the analyzed articles. We removed from the analysis stop words (e.g. and, is, the, we) and we merged singular and plural words (e.g. risk and risks). To further analyze the topics covered by the selected articles, we performed a systematic analysis of each paper to identify (1) the key social and environmental factors related to the environmental inequalities covered, (2) the methodological approach used, and (3) the main focus of the study. The social and environmental factors were identified based on a thorough read of the methodological and result section of each paper, identifying all the social, economic and environmental variables being assessed. For those articles with no explicit methodological section, the variables were identified based on the focus and repetition of words in the text. Social factors were categorized into four groups based on if the studies were differentiating population by income measures (i.e. income level), social status or measures of socioeconomic level not directly linked to income (i.e. socioeconomic level), their relative

vulnerability to risks and hazards (i.e. social vulnerability), or if these populations are taken as representative examples of people that are being affected by environmental inequalities (i.e. representative groups). For the environmental factors we produced a list of environmental variables for each article and then grouped them into major umbrella topics (e.g. green spaces and urban trees into green infrastructure). The classification of the methodological approach was based on the prevailing type of data used in each paper. Those papers that used a mix of quantitative and qualitative data were classified as mixed. Finally, to identify the main focus of each article we screened the paper aiming to find the objectives statement, usually in the abstract and/or at the end of the introduction. For those articles without an explicit objective section, we described the main focus based on our interpretation of the main text.

3. Results

3.1. Temporal patterns

Our results show that the research topic of urban environmental inequalities is relatively new and still rather underdeveloped in Latin America, with the first article published in 2002 and only 57 articles published until 2020 (Fig. 2). Temporal trends of published articles show a slow linear increase for the analyzed period with an annual average growth of 0.412 papers published per year and a maximum of 8 papers published in the years 2018 and 2020. Average citation per article

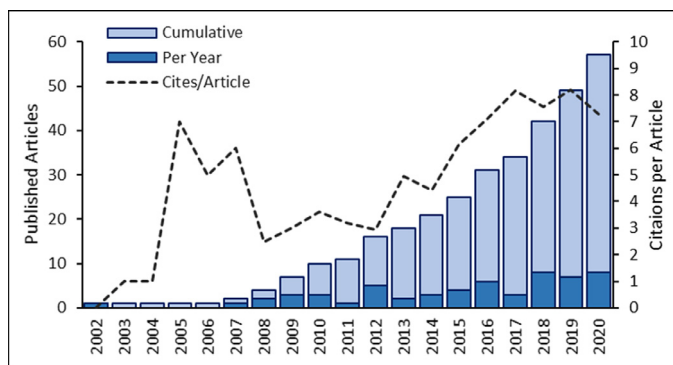


Fig. 2. Temporal trends of published articles. Cumulative and Per Year published articles on the left axis, Cites/Articles on the right axis.

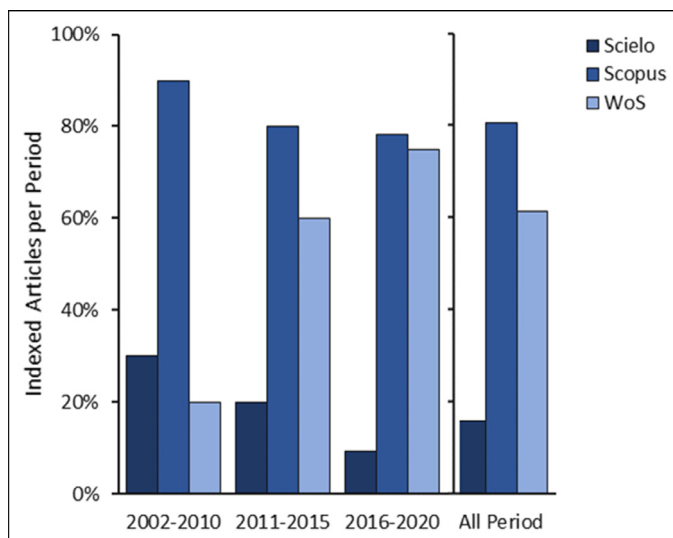


Fig. 3. Temporal trends of the percentage of articles covered by database. Columns sum more than 100% as articles can be covered by more than one database. The yearly cumulative data of indexed articles by database can be found in the supplementary material (Fig. S1).

shows a peak -due to the small number of published articles-, between 2005 and 2007, but then shows an increasing rate from 2008 to 2019. While in 2008, the average number of citations per article was 2.5, this number grew to 8.2 in 2019 (Fig. 2).

In terms of indexation of published articles, there is a clear tendency towards decreasing publications in journals indexed by Scielo and increasing proportion of articles published in journals indexed by WoS (Figs. 3, S1). While in the period 2002-2010, Scielo covered 30% of published articles, by 2016-2020 this number decreased to only 9%. On the contrary, in the same time span, WoS-indexed articles grew from 20% to 75%. Interestingly, the proportion of articles published in journals indexed by Scopus has remained high, with a slight decrease from 90% to 78% during the analyzed period ranges. Taking the entire analyzed period (2002-2020), Scopus is the database covering the largest proportion of published papers (81%), followed by WoS (61%) and by a long distance by Scielo, with only 16% of papers published in journals covered by this database (Fig. 3).

In relation to language of publications, articles published in local languages (Spanish and Portuguese) have decreased during the analyzed period, giving higher prevalence to English-written articles (Figs. 4, S2). While in the 2002-2010 period, 55% of papers were published in Spanish or Portuguese, this proportion halved to 28% in the 2016-2020 period. However, not all the remaining papers have been published in En-

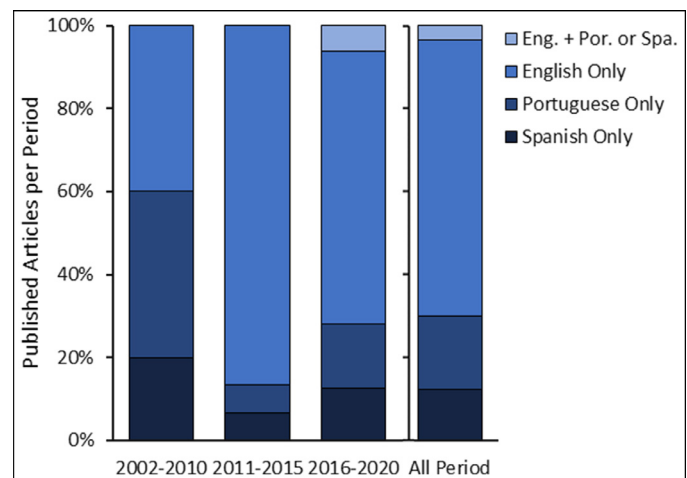


Fig. 4. Temporal trends of the percentage of articles written in each language. “Eng. + Por. or Spa.” represents the articles that are written in English plus Spanish or Portuguese. The yearly cumulative data of articles by language can be found in the supplementary material (Fig. S2).

glish only, as 6% of published paper during the 2016-2020 period were published in English plus other Latin-American language. Nevertheless, local languages still represent an important proportion of the knowledge published in the topic of urban environmental inequalities in Latin America, as 29% of all publishes articles in this topic are available only in Spanish or Portuguese (Fig. 4).

3.2. Geographical patterns

Country affiliation of the corresponding author of published articles shows a balance between institutions in Latin-American and institutions elsewhere, with a slight larger number of papers (30 of 57) associated to institutions located in Latin America (Fig. 5). Interestingly, all affiliations outside Latin-American are from countries located in North America and Europe (14 and 13 published articles, respectively), with no articles where the corresponding author is affiliated with institutions in Africa, Asia or Oceania. Brazil is by far the Latin-American country with the largest production of articles, with 16 papers, followed by Chile and Colombia, with 6 and 3 articles, respectively. Outside Latin America, United States tops the number of publications, with 13 articles, followed by United Kingdom, with 5 papers, and Finland and Spain, with 3 published articles each (Fig. 5).

In relation to the geographical location of studies, only nine out of the 19 Latin-American countries have been subject of studies covering urban environmental inequalities (Fig. 6). Brazil, Mexico and Chile are the countries with the largest number of studies, with 17, 16 and 11 studies, respectively. These countries are followed by Colombia (5 studies), Argentina (4 studies), Bolivia and Ecuador (2 studies), and Cuba and Peru (1 study). Thus, more than half of Latin-American countries have no studies on urban environmental inequalities. There is a total of 35 cities that have been assessed for environmental inequalities. The city with the largest number of studies is Santiago (7 studies), followed by Sao Paulo and Mexico City (5 studies), Rio de Janeiro, Bogota and Ciudad Juarez (4 studies), and Villahermosa (2 studies). The remaining 28 studied cities have been subject of only one study (Fig. 6).

3.3. Thematic patterns

Fig. 7 presents a word-cloud showing the top 30 concepts outlined in the abstracts of included studies. As is expected from the topic covered in this study, “environmental” (161 mentions) is the top concept covered by the published manuscripts, followed by “urban” and “city” (137 and 112 mentions, respectively), meaning that included studies are

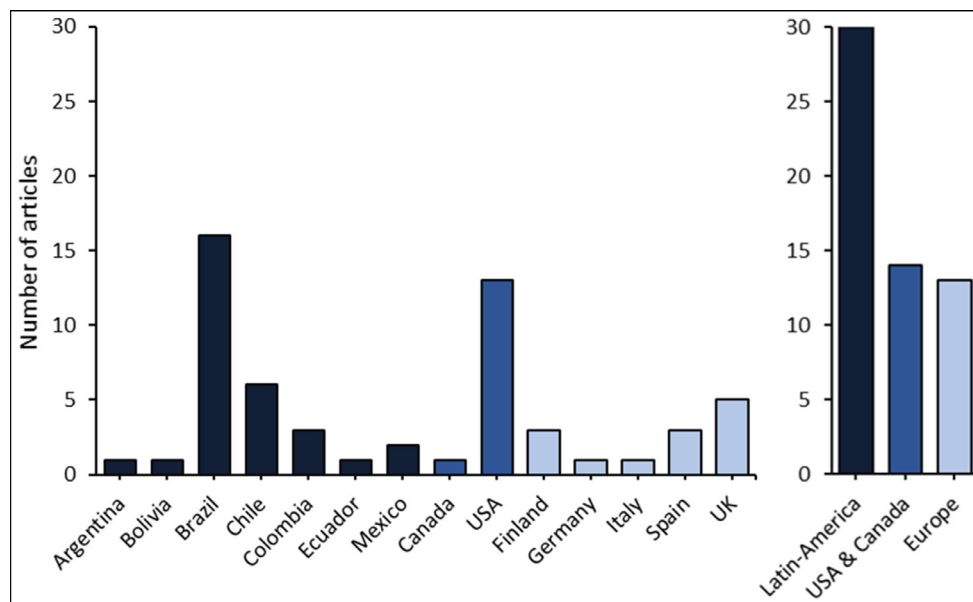


Fig. 5. Number of articles per country (or Region) of the institution of the corresponding author.

effectively covering the urban environment. Interestingly, the concepts of “inequalities” and “justice” are equally prevalent, with 50 mentions each, almost doubling the concept of “injustice” (26 mentions). The terms “social”, “socioeconomic”, “health” and “income” are the main concepts that can be related to the human dimension of covered studies, while “risk”, “pollution”, “air”, “green” and “quality”, the main concepts associated with the environmental dimension (Fig. 7). Other prevailing terms are “areas”, “spatial”, “distribution” and “exposure”, which together with “study”, “analysis” and “data”, relate to the type of research covered by these studies. Finally, other frequent words included in the abstract, such as “population” and “residents”, can be related to the subject of studies.

Table 1 shows a categorized summary of each of the studies covered in this review, including the studied country and city, the social and environmental factors related to inequalities, the research type and the main focus of research. In relation to the social dimension, the large proportion of studies (72%) uses some measure to classify different groups of people that are subject to environmental inequalities. Based on our analysis (see methodology section), the most prevalent measure of social difference is based on social vulnerability, followed by socioeconomic level and by income level (17, 15 and 9 articles respectively). The remaining studies (16 articles) do not use a measure of social differentiation but focus on a specific representative social group or groups that are subjected to environmental inequalities (Fig. 8). Regarding to the environmental dimension, green infrastructure and air pollution are the most frequent environmental factors related to inequalities, with 14 and 11 articles focused on these elements respectively. Environmental risks, industrial hazards, environmental quality and water resources (10, 9, 7 and 6 articles, respectively) are also relevant subjects of research. While less predominant, other environmental factors covered by the studies include urban waste, urban heat and health-related environmental hazards (Fig. 8). With respect to the methodological approaches, 34 of the studies uses quantitative methods, 19 qualitative, and 4 a mix of quantitative and qualitative methods (Fig. 8).

4. Discussion

Social and economic inequalities are prevalent and well-known issues of Latin-American cities [14,84,92]. Nevertheless, it seems that the relationship between these urban inequalities and environmental factors have only recently started to receive attention by the scientific community [35,56]. Our results support these conclusions. Indeed, while we set

our search to start in the year 1980, the first record we found covering the topic of urban environmental inequality (or justice) was published only by 2002. However, our data show an increasing number of publication and citations during the last decade, suggesting that urban environmental inequalities is still an emerging topic, but is gaining relevance within the Latin-American Region. Similar trends for published papers have been observed for other related urban environmental topics, such as urban ecosystem services [75], urban forestry [64] and urban green infrastructure [11], highlighting the increasing importance that environmental factors are taken within urban areas of developing regions, such as Latin America [95].

There is evidence pointing to the potential biased outcomes resulting from reviews based only in a single or a few predominant global databases, such as Scopus and WoS [17,88,89]. While we did find articles covered exclusively by Scielo, these represented only 16% of selected articles, proportion that decreased to 9% by the end of the studied period. Thus, the large majority of articles about urban environmental inequalities published in mainstream Latin-American Journals are also covered by the predominant international databases, which does not support an indexation bias. The growing representation of published papers in international databases could be the result of an increasing number of journals indexed by these databases, specifically by WoS after the inclusion of the *Scielo Citation Index* in 2014 [70], but also because the academic incentives of Latin-American researchers to publish in WoS or Scopus indexed journals [60]. In addition, we also found a temporal tendency towards increasing publications in English-written journals, which could be associated to the incentive of publishing in these journals due to the relative higher citation outputs reached by articles written in English instead of local languages [26]. Even though leaving the local languages could be seen as a factor that reduces the impact of studies outcomes in the local scientific community [52], it also suggests that studies on urban environmental inequalities in Latin American have reached a maturity to start moving from targeting a local audience, towards reaching a broader international community, therefore increasing its contribution to the scientific discipline.

An interesting, however not surprising geographical pattern was the large contribution of corresponding authors located in European and North American institution, which summed-up reach 47% of the analyzed papers. There is increasing awareness of some research practices that have been coined as “helicopter or parachute research”, which relate to scientists from richer developed countries carrying out studies with data gathered from poorer undeveloped countries with almost no

Table 1
Detailed summary of the 57 studies covered by this systematic review.

Article	Country	City	Social Factor	Environmental Factor	Methods	Article Main Focus
Boso et al [10]	Chile	Coyhaique	Social vulnerability	Air pollution	Mixed	Residents' perception about air pollution and associated health risks
Chavez-Rodriguez et al [18]	Mexico	Monterrey	Representative groups	Environmental quality	Qualitative	Discourses and narratives on urban mobility reinforcing environmental injustice issues
Cruz-Sandoval et al [21]	Mexico	Guadalajara	Socioeconomic level	Green infrastructure	Quantitative	Spatial association between urban trees and socioeconomic status
Lome-Hurtado et al [51]	Mexico	Mexico City	Social vulnerability	Air pollution	Quantitative	Spatial association between vulnerable groups and air pollution concentration
Mahady et al [53]	Mexico	Mexico City	Social vulnerability	Air pollution	Quantitative	Environmental justice mapping index for clean transportation interventions
Quimbayo Ruiz et al [69]	Colombia	Bogota	Representative groups	Environmental quality	Qualitative	Strategies and practices to cope with urbanization impacts
Rivera-Florez et al [73]	Colombia	Medellin	Representative groups	Environmental risks	Qualitative	Formulation of a community risk management and disaster prevention plan
Tittor & Toledo López [90]	Argentina	Córdoba	Representative groups	Industrial hazards	Qualitative	Narratives of a protest movement that emerged in response to a bioethanol production plant
Anguelovski et al [6]	Colombia	Medellin	Representative groups	Green infrastructure	Qualitative	Green infrastructure interventions as driver of socio-spatial inequities
Bosisio & Moreno-Jiménez [9]	Argentina	Santa Fe	Social vulnerability	Environmental risks	Quantitative	Unequal exposure of vulnerable and deprived social groups to two flooding events
Ribeiro [72]	Brazil	Sao Paulo	Socioeconomic level	Air pollution	Quantitative	Association between traffic related air pollution and respiratory cancer by socioeconomic status
Silva & Samora [83]	Brazil	Campinas	Representative groups	Water resources	Mixed	Impact of drought on urban residents and their perceptions on the management of water crisis
Rodríguez-Guerra & Cuvi [74]	Ecuador	Quito	Socioeconomic level	Air pollution	Quantitative	Association between fine and coarse particulate matter and socioeconomic-based land price
Rubiano [77]	Colombia	Bogotá	Socioeconomic level	Green infrastructure	Quantitative	Distribution of tree coverage and greenness, and its spatial association with socioeconomic strata
dos Santos [79]	Brazil	Curitiba	Income level	Green Infrastructure	Quantitative	Association between built environment, income level and physical activity at school surroundings
Albuquerque Sant'Anna [3]	Brazil	Rio de Janeiro	Social vulnerability	Environmental risks	Quantitative	Effects of public policies on the occurrence of natural disasters related to extreme rainfall
Amorim & Xavier [5]	Brazil	Sao Paulo	Income level	Environmental quality	Qualitative	Reflection on the relationship between city and nature in the context of socio-spatial inequalities
Fernández & Wu [33]	Chile	Santiago	Social vulnerability	Environmental quality	Quantitative	GIS-based framework for identifying priority areas for urban environmental inequity mitigation
Nygren [62]	Mexico	Villahermosa	Socioeconomic level	Environmental risks, Green infrastructure, Urban Waste, Water Resources	Qualitative	Links of environmental inequalities with urban governance and justice structures
Nygren & Wayessa [63]	Mexico	Villahermosa	Representative groups	Environmental risks	Qualitative	Politics of displacements and socially differentiated experiences of environmental injustices
Orta Ortiz & Geneletti [65]	Cuba	Havana	Social vulnerability	Green infrastructure	Quantitative	Mismatches between ecosystem services flows and demand
Duarte et al [28]	Brazil	Urban areas in general	Income level	Green infrastructure	Qualitative	Historical process of urban greening in Brazil, inequalities and current challenges to move forward
Valdivia [94]	Ecuador	Esmeraldas	Representative groups	Industrial hazards	Qualitative	Political ecology analysis of the relation between urban residents and an oil refinery
Schonardie [80]	Brazil	Urban areas in general	Representative groups	Environmental quality	Qualitative	Theoretical reflection on human rights, environmental justice and sustainable cities
Fernández-Álvarez [31]	Mexico	Mexico City	Socioeconomic level	Green infrastructure	Mixed	Association between the distribution of green public spaces and lower socioeconomic sectors
Guibrunet et al [40]	Mexico, Chile	Mexico City, Santiago	Representative groups	Urban Waste	Qualitative	Urban metabolism of residential waste management in local neighborhoods
Castillo-Gallardo [16]	Chile	Arica	Representative groups	Industrial hazards	Qualitative	Cultural construction of an inequality concept from the experience of living in a heavy-metal polluted site
de la Barrera et al [24]	Chile	Santiago	Income level	Green Infrastructure	Quantitative	Indicators for evaluating and planning green spaces within urban areas

(continued on next page)

Table 1 (continued)

Article	Country	City	Social Factor	Environmental Factor	Methods	Article Main Focus
Fernández & Wu [32]	Chile	Santiago	Socioeconomic level	Air pollution, Green Infrastructure, Heat	Quantitative	Spatial association between environmental and socioeconomic variables at multiple scales
Bellini et al [7]	Brazil	Rio das Ostras	Income level	Environmental quality	Quantitative	Distribution of water, sewage, waste and vegetation cover and its association with income and literacy level
Ioris [46]	Peru	Lima	Income level	Water resources	Qualitative	Territorialization of water scarcity from interactions between government, economic sectors and civil society
Gurgatz et al [41]	Brazil	Paranaguá	Income level	Air pollution	Quantitative	Methodological approach for using tree bark to estimate relative air pollution risk
da Silva Rosa et al [22]	Brazil	Niterói, Jab. Guararapes	Representative groups	Environmental risks	Qualitative	Environmental education projects in landslide-risk areas of historically socio-environmental vulnerability
Escobedo et al [29]	Colombia	Bogotá	Socioeconomic level	Green infrastructure	Quantitative	Spatial association between ecosystem services provided by trees and socioeconomic strata
Grineski et al [38]	Mexico	Tijuana	Social vulnerability	Industrial hazards	Quantitative	Association between the distribution of industrial parks and neighborhoods' sociodemographic
Sandoval et al [78]	Chile	Chaitén	Representative groups	Environmental risks	Qualitative	Asymmetric relationship in managing residents relocation and recovery after volcanic eruption
Gonçalves et al [34]	Brazil	Porto Velho	Social vulnerability	Environmental hazards	Quantitative	Socio-environmental vulnerability in relation to forest fires and related health effects
Habermann et al [42]	Brazil	Sao Paulo	Socioeconomic level	Air pollution	Quantitative	Association between exposure to vehicle-related air pollution and socioeconomic status
Morales et al [25]	Argentina	Buenos Aires	Representative groups	Urban Waste	Qualitative	Urban residents' imaginaries of sanitation technologies for managing house-level sewage
Hetrick et al [44]	Brazil	Altamira	Socioeconomic level	Green infrastructure	Quantitative	Distribution of urban vegetation and its association with social indicators
Romero-Lankao et al [76]	Colombia, Mexico, Chile	Bogotá, Mexico City, Santiago	Social vulnerability	Air pollution, Heat	Quantitative	Association between air pollution and temperature-related health risk, and socioeconomic vulnerability
Bustamante & Médiu [13]	Bolivia	Cochabamba	Representative groups	Water resources	Qualitative	Revision of elements to understand the current access to and use of water in the southern part of the city
Domínguez-Cortinas et al [27]	Mexico	Mexico City	Social vulnerability	Environmental hazards	Quantitative	Cross sectional study on the relation of health issues with social vulnerability and environmental hazards
Grineski et al [39]	Mexico	Ciudad Juarez	Social vulnerability	Air Pollution, Environmental risks, Heat	Quantitative	Spatial association between climate change-related hazards and socio-demographic variables
McDonald & Grineski [57]	México	Ciudad Juarez	Social vulnerability	Water resources	Quantitative	Spatial association between plumbing coverage and sociodemographic variables
Wright Wendel et al [97]	Bolivia	Santa Cruz	Income level	Green Infrastructure	Mixed	Perceptions, preferences, and accessibility of green spaces
Urkidi & Walter [93]	Argentina	Esquél	Representative groups	Industrial hazards	Qualitative	Connections between Latin-American anti-mining movements and the environmental justice framework
de Freitas Vilas Boas Gomes [23]	Brazil	Guarapuava	Income level	Environmental quality	Quantitative	Association between environmental quality and income level distribution
Grineski & Collins [37]	México	Ciudad Juarez	Social vulnerability	Industrial hazards	Quantitative	Spatial association between marginalized neighborhoods and the presence of industrial facilities
Reyes & Figueroa [71]	Chile	Santiago	Socioeconomic level	Green infrastructure	Quantitative	Distribution size and accessibility of green areas and its relationship with neighborhood socioeconomics
Cartier et al [15]	Brazil	Rio de Janeiro	Social vulnerability	Industrial hazards	Quantitative	Spatial association between distance to industrial area and socio-demographic indicators
Lara-Valencia et al [49]	Mexico	Nogales	Socioeconomic level	Industrial hazards	Quantitative	Spatial association between hazardous-waste facilities and socioeconomic groups
Vázquez & Salgado [96]	Chile	Santiago, San Pedro	Socioeconomic level	Environmental risks	Quantitative	Spatial association between environmental risks and neighborhood socioeconomic level
Grineski & Collins [36]	México	Ciudad Juarez	Social vulnerability	Industrial hazards	Quantitative	Spatial association between industrial facilities and residential sociodemographic characteristics
Stevens et al [85]	Mexico	Country level	Socioeconomic level	Air pollution, Water resources	Quantitative	Disparities in mortality caused by environmental factors depending on municipal socioeconomic status
Alves [4]	Brazil	Sao Paulo	Socioeconomic level	Environmental risks	Quantitative	Association between exposure to hydrological risks and groups socioeconomic conditions
Chiesa et al [19]	Brazil	Sao Paulo	Social vulnerability	Environmental hazards	Quantitative	Distribution of environmental and social factors related to increased vulnerability to respiratory diseases

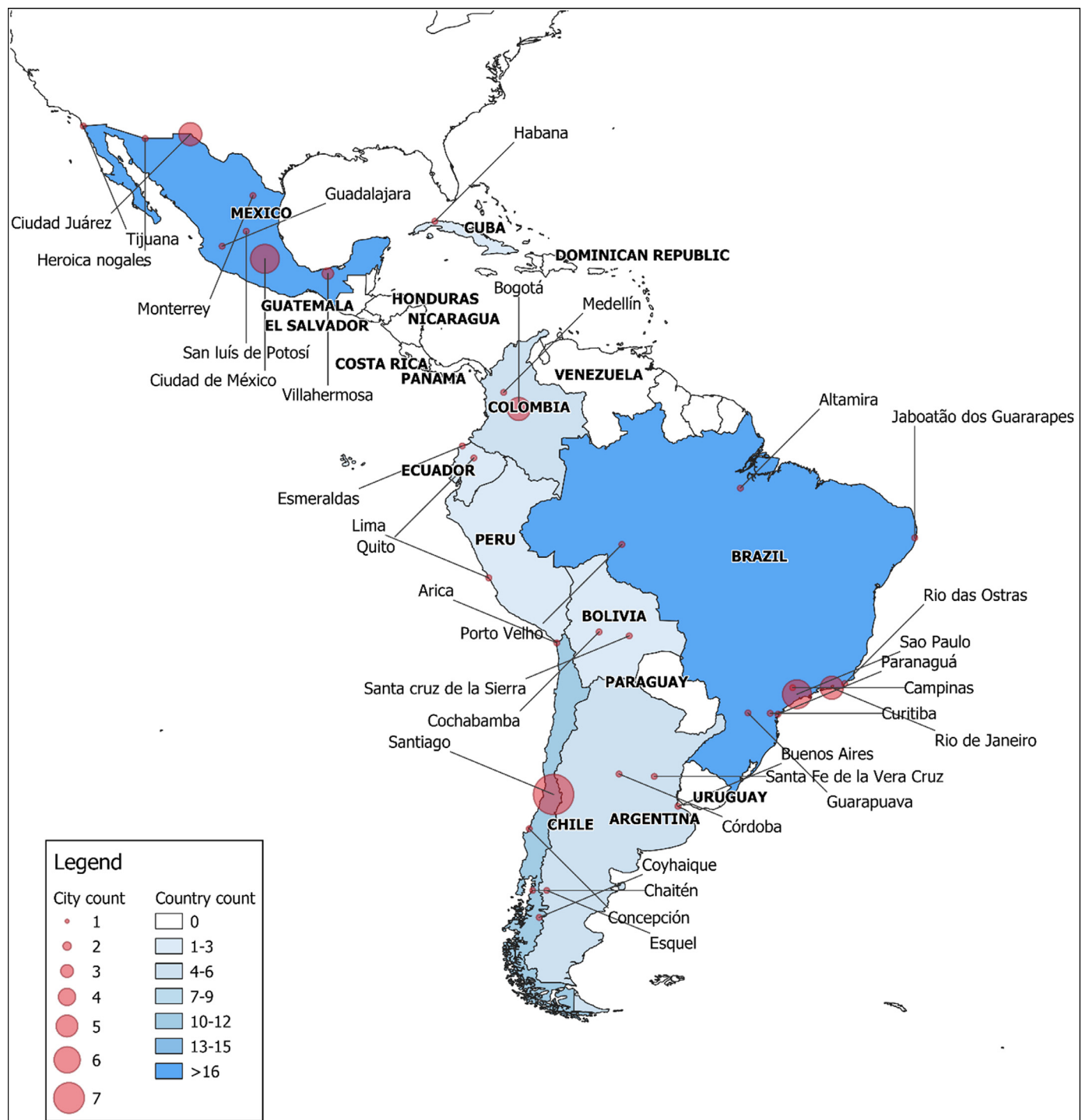


Fig. 6. Distribution of published studies per country and city.

collaboration with local scientist, and no retribution to the local community that made possible the studies [43]. However, we do not have evidence for this regarding the topic of urban environmental inequalities. On the contrary, based in our own experience and previous studies [8], we think that many of these international affiliations are mostly associated to Latin-American researchers doing their graduate studies or working on developed countries and also to foreign scholars performing long-term collaborative research in Latin America. This is aligned with the increasing dominance of publication in foreign English-written journals, which highlights the increasing contribution that Latin-American scholars are generating on this topic.

In relation to the geographical distribution of studies, we found a clear bias towards a few countries and cities. Brazil and Mexico concentrate more than half of all studied areas, which could be expected, as these countries concentrates more than half of the urban population of the Latin-American region [91]. Nevertheless, Chile, with less than 4% of the urban population of the region, concentrates close to 18% of the studied areas, meaning that the geographical bias cannot be explained only because the size of urban population. In fact, we found that more than half of Latin-American countries have not been studied so far, even though most of them are known for presenting cities with important levels of spatial segregation and economic inequalities [92].



Fig. 7. Wordcloud of the main topics covered in the abstracts of the analyzed studies.

Similar biased patterns have been found for studies on urban ecosystem services [75] and urban forestry [64], implying that this bias is not unique to urban environmental inequalities, but probably a more generalized pattern associated to the unequal development of scientific research between countries [20]. Furthermore, this geographical bias is also seen within countries, with major cities receiving considerably more attention. In fact, the six more studied cities concentrated more than 50% of all the studies carried out in the region. These results are of great concern because many relevant aspects of urban environmental inequalities in Latin America are not being studied or reported, generating a gap of knowledge for local decision-making, but also for advancing the discipline of urban environmental inequalities.

Results from the thematic patterns analysis confirm that selected studies are effectively covering the topic of urban environmental inequalities, providing a more in-depth perspective of the specific focus and the social and environmental factors analyzed by the studies carried-out in Latin America. For example, while in North America the ethnic and racial dimensions have been a prevalent social factor associated to environmental inequalities [2,12,59], our results show that in Latin America these factors are almost absent from the literature, and the majority of the studies focus primarily on the interlink between the social and economic dimensions. This difference could be related, among other factors, on the predominant focus that urban inequalities studies on Latin America have had on social differentiation by income level [92], on the limited and inconsistent data of race and ethnicity within

the region [87], on the weak explanatory power of race and ethnicity for explaining social inequalities in this region [67], and on cultural perceptions on who the society and researchers identify as the minorities potentially suffering unfair environmental conditions. On the contrary, the environmental factors are more diverse, including air and water pollution, industrial and environmental hazards, environmental risk and quality, and green infrastructure, which have also been predominant themes covered by the urban environmental inequality studies in developed regions [1]. This reveals how different geographical regions and countries may approach urban environmental inequalities depending on their specific characteristics and historical background, highlighting the relevance of promoting and embracing research on developing regions to reduce potential knowledge bias and advance the discipline.

Even though we do not have a comparison with other regions, is interesting that in relation to the methodological approaches, while quantitative methods tend to dominate, a third of the analyzed studies uses qualitative methods, suggesting that both approaches are playing important roles in advancing the topic in the region. Similar patterns were found in a literature review by Mayorga & Vázquez [56]. The dominance of the quantitative approaches can be linked to the prevalence of studies assessing inequalities through spatial analysis, and therefore more associated to the concept of “urban environmental inequalities”. At the other hand, most of the studies using qualitative approaches are based on case studies or previous evidence, assuming that these representative groups have received an unfair treatment in relation to some environmental factor, and therefore more related to the concept of “urban environmental justice”. We consider this mixture of approaches as a good outcome, as it generates a diversity of knowledge that help producing a more integrated vision of urban environmental inequalities in Latin America.

5. Conclusion

Although still relatively young, urban environmental inequalities research in Latin America has shown a relevant increase, both in published articles and number of citations, during the last two decades. The research has also been increasingly internationalized during the last decade, by transitioning from papers written in local languages and published in journals covered by regional databases, to articles written in English and published in journals covered by international database. These, together with the diversity of approaches and covered topics, suggests that the discipline is rapidly reaching a maturity in the region, and could become a relevant source of knowledge for advancing the science. Nevertheless, there is still a large research bias towards a few countries that concentrate the large proportion of published studies, leaving a gap of knowledge regarding the specifics of urban environmental in-

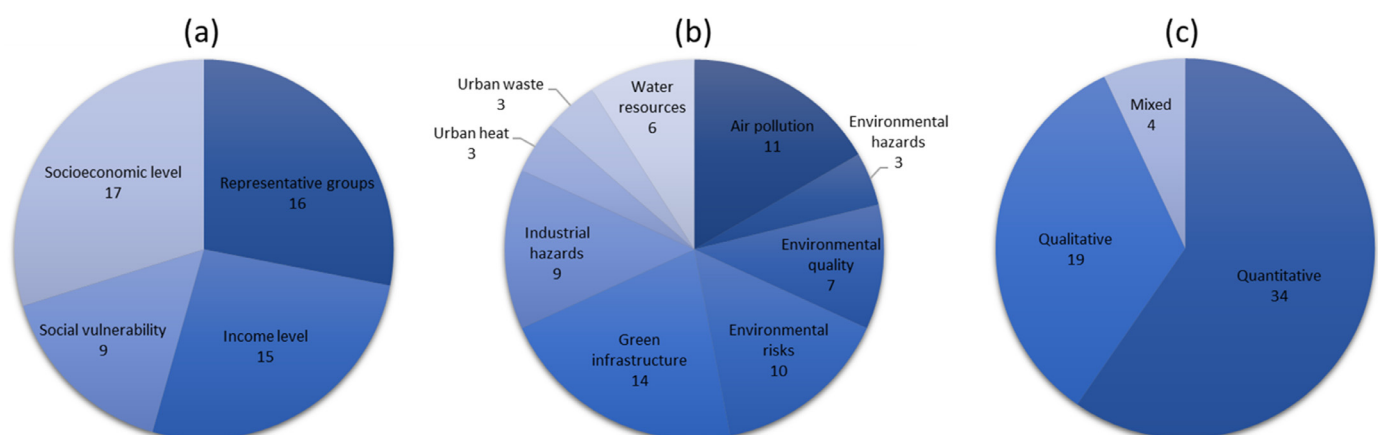


Fig. 8. Pie-charts representing the relative proportion of (a) social factors, (b) environmental factors and (c) methodological approaches associated to the analyzed studies. The data for building the charts is based on [Table 1](#).

equalities in the other countries. Certainly, there could be other sources of research (e.g. grey literature, not indexed journals, other scientific databases) not covered by this review that may have relevant information to help filling this gap, but unfortunately that type of information will hardly reach the international scientific community if not written in English and published in mainstream international journals. Therefore, we hope that this review helps to make visible to a wider audience those relevant articles that can have gone unnoticed because database and language barriers, and also help promoting the advance of the discipline in the Latin-American region, in particular, in those countries where the research on urban environmental inequalities seems to be absent. This could lead to key novel insights for advancing our understanding on the factors driven the inequalities, which can be fundamental for developing better solutions to address the issues of urban environmental inequalities in Latin America or elsewhere.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.wds.2023.100055](https://doi.org/10.1016/j.wds.2023.100055).

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